



Engineered porous borophene with tunable anisotropic properties

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ABSTRACT

Monolayer borophene, a two-dimensional nanostructure, has received much attention due to its high flexibility, excellent elasticity, high strength, and interesting electronic characteristics. However, borophene is highly anisotropic, and therefore programming their anisotropic thermo-mechanical properties is of great interest for a wide range of applications in electronic devices, ion batteries, hydrogen storages, and supercapacitors. To determine the impact of porosity and pore topology on thermo-mechanical properties of porous borophene, non-equilibrium molecular dynamics simulation and finite element approach are adopted. It is found that pore engineering not only effectively controls the thermo-mechanical properties of borophenes but also can offer a strategy to transform an anisotropic borophene to a square quasi-isotropic one. It is shown that the lattice thermal conductivity and mechanical properties along armchair and zigzag directions of monolayer borophenes can be tuned and equalized by changing the pore nanoarchitecture. This work unveils the potential of pore engineering in multiple length scales for tuning the anisotropic thermo-mechanical properties of monolayer borophenes and 2D nanomaterials, such as phosphorene, to develop nanodevices with a desired multifunctional performance.

1. Introduction

Two-dimensional (2D) materials are one of the most prominent recent advanced nanomaterials due to their potential applications in the next generation of electronic devices and biomedical systems [1–3]. Most recently, a new class of 2D atom-thin sheets known as borophene (crystalline 2D boron sheets) has been synthesized on Ag (111) substrates [4]. So far, four different phases of borophene have been experimentally synthesized (2-Pmmn, β 12, χ 3, and honeycomb phases) [5–7]. The 2-Pmmn phase of borophene exhibits remarkable anisotropic mechanical, electronic, optical, and thermal properties due to its anisotropic atomic structure [4,8]. Furthermore, borophene presents a buckled morphology and metallic characteristics, making borophene a great material for energy storage systems [9,10]. Inherent to its low mass density ($\sim 5.35 \times 10^{-7}$ kg/m²) [11], excellent tensile strength and intrinsic phonon-mediated superconductivity [4,11], borophene offers an advanced alternative nanomaterial for flexible nanoelectronics and near-visible plasmonics [8,12–14]. Moreover, the storage capacity of borophene is far higher than the graphite electrode and is the highest

among other 2D materials, making borophene a suitable anode material for ion batteries [13,15,16]. Borophene also exhibits biocompatibility and promising bioimaging performance [17].

Due to the short B–B bonds and the low mass of boron, the 2-Pmmn phase of borophene has outstanding mechanical characteristics. An experimental study has reported in-plane elastic modulus of borophene as 586.2 GPa and 1372.4 GPa with a corresponding thickness of 0.27–0.31 nm along the zigzag and armchair directions (Fig. 1a), respectively [4], compared to graphene that reveals elastic modulus of about 1000 GPa and possesses a thickness around 0.34 nm [18]. The ultimate tensile strength of borophene has been reported ~ 46.5 GPa and ~ 78.6 GPa along the zigzag and armchair directions [19,20], while the tensile strength of graphene is ~ 123.5 GPa [18]. Because of borophene negative in-plane Poisson's ratio (~ -0.04 along armchair and ~ -0.02 along zigzag), boron nanosheets show high mechanical flexibility along the armchair direction that is an exciting feature among 2D nanostructures for the fabrication of flexible nanodevices and nanoelectronics [21–23]. Moreover, the 2-Pmmn phase of borophene shows a high stiffness-to-weight ratio (~ 517 m² s⁻²) among the other 2D

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nanomaterials, including graphene ($\sim 453 \text{ m}^2 \text{ s}^{-2}$) [24].

Borophene also possesses unique thermal properties. Its strong phonon-phonon scattering results in an unexpectedly low lattice thermal conductivity. The lattice thermal conductivity of borophene is found to be $\sim 76 \text{ W/mK}$ along the *zigzag* and $\sim 147 \text{ W/mK}$ along the *armchair* directions [25], at room temperature, which is much smaller than graphene with a thermal conductivity $\sim 5300 \text{ W/mK}$ [26]. Furthermore, borophene has a significant negative thermal expansion coefficient $\sim -6.9 \times 10^{-6} \text{ 1/K}$ along the *zigzag* and $\sim -79 \times 10^{-6} \text{ 1/K}$ along the *armchair* directions [27]. The thermal conductivity anisotropy of borophenes can be promising for programming the heat-flow pathways and thermal management of nanoelectronic devices.

Porous 2D nanomaterials have been utilized in a series of applications, including energy generation and energy storage devices. The morphology of pores can provide a higher transport rate for ions in lithium-ion batteries [28] or the electrolyte of supercapacitors [29]. Furthermore, flexible and stretchable electronics, such as electronic skins and epidermal electronics, require specific heat conduction and mechanical properties [30]. The anisotropy engineering of 2D nanomaterials has a potentials to also improve biosystems by directing the desired mechanical properties and heat dissipation [31,32]. Pore engineering can have a significant impact on the functionality and reliability of the flexible and stretchable devices that require specific thermo-mechanical properties.

Recent advances in nanotechnology and advanced manufacturing have enabled controlling and engineering porosity and defects in the nanostructure of manmade materials, including 2D and 3D nanomaterials [33]. The plasma irradiation [34], bottom-up [35], and laser scribing [36] techniques offer the design and engineering of the pores with diameters less than a nanometer for 2D nanomaterials (e.g., graphene). Experimental studies on borophenes demonstrated the low required energy for the formation of vacancies in their nanostructures. Hence, vacancies are anticipated to be formed during the fabrication process of borophenes [4,37] deliberately (*engineered pore*) or unintentionally (*defect*).

Recently, a few studies have attempted to tune the anisotropic transport characteristics of borophene, which leads to enhanced borophene performance in electronic and optical nanodevices. It has been shown that the anisotropic current ratio in the *armchair* and *zigzag* directions could be remarkably tuned by oxidation strategy and applying strain in a pristine borophene [38,39]. However, tuning the anisotropic thermal and mechanical properties of borophene has not yet been studied specifically through pore engineering, a strategy that has recently been shown to be effective for tuning the thermal and mechanical properties of other nanomaterials, e.g., graphene and phosphorene [40–42].

This paper conducts a series of non-equilibrium molecular dynamics (NEMD) simulation to provide insight into the effect of pore topology on the thermo-mechanical properties of the 2-Pmmn phase of porous borophenes. The 2D profile of temperature and stress distribution is determined, and correlations between lattice thermal conductivity and mechanical properties with pore architecture are proposed. We explore the pore engineering approach to investigate the possibility of transforming the anisotropic thermo-mechanical behavior of borophene to a square quasi-isotropic one. By comparing the numerical results of NEMD simulation and finite element modeling (FEM), we investigate the similarity between the variation of thermo-mechanical properties of nano-scale porous borophene sheets and their macroscopic counterparts with the same engineered microstructure.

2. Methodology

In the present study, LAMMPS package is used for performing molecular dynamics calculations [43]. As the accuracy of estimated properties by molecular dynamics simulation mainly depends on the precision of potential functions used, we have employed ReaxFF potential function [44] to describe atomic interaction in the borophene sheet to allow reactive interactions and atomic bond formation or breakage [45]. Borophene films with periodic boundary conditions are used in traverse direction to eliminate the effects of atoms at the boundary edges. Moreover, periodic boundary conditions also minimize the finite length effects. All simulations are carried out at an average temperature of 300K, and the time increment of 1 fs.

We first study the mechanical properties of the porous borophene nanostructure. Before applying a tensile load with, the nanostructures are relaxed using a Nose-Hoover barostat and thermostat (NPT) for 2.5 ps. The relaxation process minimizes potential energy and vanishes unwanted stress in the nanostructure. To calculate mechanical properties (Elastic modulus, toughness, fracture stress, and fracture strain), we define the strain as $\epsilon = \frac{\Delta L}{L_0}$ where, L_0 represents the initial length (30 nm), and ΔL denotes the change in length caused by applying an external strain to the borophene sheets. The length of the simulation box, along the loading direction, is increased by a constant engineering strain rate of $1 \times 10^9 \text{ s}^{-1}$. To accurately ensure uniaxial stress conditions, the periodic simulation box perpendicular to the tensile direction is altered using the NPT method. The uniaxial stress-strain curves include an initial linear domain, followed by a nonlinear behavior up to the ultimate tensile strength, where the maximum load-bearing capacity of the borophene sheets is reached. Further increase of strain beyond the ultimate tensile strength point leads to a sudden drop in the stress values followed by the ultimate rupture in the borophene sheets. The slope of the linear part of the uniaxial stress-strain curve is reported here as

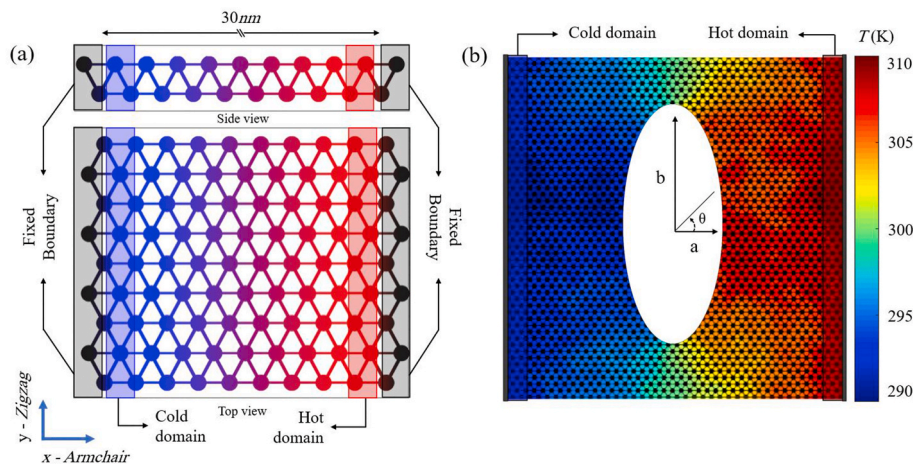


Fig. 1. (a) System setup for NEMD thermal conductivity calculations and (b) 2D temperature distribution of porous borophene calculated by NEMD.

elastic modulus. The area under the stress-strain curve indicates the toughness of the borophene sheets. To calculate the mechanical and thermal properties, we assumed thickness of 0.29 nm for the borophene sheets.

The stress components (S) are calculated by Virial theorem [46,47]:

$$S = \frac{1}{V} \sum_{a \in V} \left[-m \vec{v}_a \otimes \vec{v}_a + \frac{1}{2} \sum_{a \neq b} (\vec{r}_{ab} \otimes \vec{F}_{ab}) \right] \quad (1)$$

where m and $\vec{v}_a$ represent mass and velocity vector of the atom a , respectively, $\vec{r}_{ab} = \vec{r}_b - \vec{r}_a$ is the position vector of the atom a with respect to atom b , $\vec{F}_{ab}$ denotes the force vector on atom a caused by atom b , V represents the volume of the borophene nanostructure, and $\otimes$ denotes the Tensor product. Afterward, 2D von Mises stress distribution is calculated by dividing the sample into 20 subdomains in each direction ($20 \times 20 \times 1$ subdomains).

The nanoarchitecture of defected borophene and molecular dynamics setup for the evaluation of lattice thermal conductivity are shown in Fig. 1. A few rows of boron atoms at the two ends of the x -direction of borophene are fixed to prevent the slippage of atoms. Periodic boundary conditions are assumed in *armchair* direction for the calculation of thermal conductivity in *zigzag* direction (K_y). As shown in Fig. 1, hot and cold domains have been located beside the fixed boundaries. For calculation of thermal conductivity in *armchair* direction (K_x), a similar procedure is repeated by changing the direction of temperature gradient and periodic boundary condition.

The lattice thermal conductivity of porous borophene nanostructure along in-plane directions is calculated using the non-equilibrium molecular dynamics (NEMD) simulation [48–50]. The porous borophene has been relaxed using NPT ensemble at room temperature for 2.5 ps. After obtaining the equilibrated nanostructure, a temperature gradient imposed by the presence of hot and cold domains generates a heat flux within the porous borophene nanosheet. To calculate average energy added to and subtracted from the borophene, a microcanonical ensemble (NVE) is applied for 1 ns (Fig. S1 in Supporting Information).

We finally calculate the effective thermal conductivity (K_{eff}) of the porous borophene using the Fourier heat conduction equation [49]:

$$K_{eff} = q \frac{L}{\Delta T} \quad (2)$$

where q is heat flux between the cold and hot domains, $L = 30$ nm represents the length of the borophene nanosheet, and ΔT is the temperature difference between the two ends.

To obtain 2D temperature distribution (Fig. 1b), the local temperature is calculated based on the total kinetic energy of the group of atoms:

$$T = kE \left(\frac{3}{2} N k_B \right)^{-1}; \quad kE = \sum_i \frac{1}{2} m v_i^2 \quad (3)$$

where T represents local temperature, k_B is the Boltzmann constant, N is the number of atoms at each subdomain, KE represents total kinetic energy, and m and v represent the mass and velocity of each atom.

To explore the similarity between the effective thermo-mechanical properties of porous borophene predicted by atomistic and continuum approaches, FEM (in addition to NEMD) is used for analysis. The constitutive properties of the developed nanostructure for the FEM analysis stem from the calculated properties of pristine borophenes using NEMD simulation.

3. Results and discussion

3.1. Validation

To validate our NEMD simulation results, the mechanical (elastic modulus) and thermal (lattice thermal conductivity) properties of the pristine borophene are calculated and compared with those reported in the literature (Table 1). Comparing our calculated thermo-mechanical properties by NEMD simulation with those reported in previous studies (NEMD method [25] and first principles techniques [50,51]), corroborates the accuracy of our computational multiscale model. Due to the computational cost, the size of the investigated borophene is limited to 30×30 nm for predicting the properties of pristine and engineered porous borophene; the limited selected size is the reason for the deviation of our NEMD predictions for thermal conductivity to the bulk result. This discrepancy is associated with the size-dependency of the thermal properties of low-dimensional nanomaterials. When the length scale is comparable to the phonon mean free path, phonons experience limited scatterings during transport, leading to ballistic phonon transport and, consequently, a reduction in thermal conductivity [52–54].

Fig. 2 shows the length dependency of thermal conductivity along *armchair* and *zigzag* directions as well as the thermal conductivity anisotropy of pristine borophenes. The length dependency of the thermal conductivity is calculated using the following equation [55]:

$$\frac{1}{K(L)} = \frac{1}{K_\infty} \left(1 + \frac{\Lambda}{L} \right) \quad (4)$$

where K_∞ corresponds to the inherent thermal conductivity for structure with infinite length, Λ denotes an effective phonon mean free path, and L represents the sample length. By fitting equation (4) to the obtained NEMD values for different lengths, the thermal conductivity of bulk borophene is estimated as 70.4 W/mK and 140.8 W/mK for the *zigzag* and *armchair* directions, respectively. In the meantime, the effective phonon mean free path takes the values of 16.6 nm and 21.8 nm along *zigzag* and *armchair* directions, which are in a good agreement with the previously reported values of 16.7 ± 1.7 nm and 21.4 ± 1.0 nm for *zigzag* and *armchair* directions [25].

The length dependency of the mechanical properties of the pristine borophene is also investigated. As shown in Fig. 3, the elastic modulus, toughness, fracture stress, and fracture strain/stretchability are length-dependent up to $L \sim 30$ nm, while this dependence almost vanishes for greater lengths.

3.2. Thermal and mechanical analysis of porous borophene

Pore engineering has recently been introduced as a robust strategy to tune the multifunctional properties of advanced materials with rationally-designed microstructures to develop lightweight and high-performance materials and structures [56–60]. Advances in precise manufacturing and nanotechnology have enabled extending the pore engineering strategy to nanomaterials to design their nanostructure and to tune their mechanical stiffness/strength, toughness, and thermal conductivity. In this subsection, the effect of circular pores on the thermal conductivity and mechanical properties of monolayer borophene is investigated.

The effect of porosity on the thermal conductivity of borophene sheets containing periodic circular pores are shown in Fig. 4. The

Table 1

Comparison of calculated thermal conductivity and elastic modulus of pristine borophene with previous researches.

	Thermal conductivity (<i>Armchair</i>) W/mK	Thermal conductivity (<i>Zigzag</i>) W/mK	Elastic modulus (<i>Armchair</i>) GPa	Elastic modulus (<i>Zigzag</i>) GPa
Present study	85.2 ± 2.4	45.3 ± 5.2	1373.1 ± 22.7	597.2 ± 11.7
NEMD [25]	83.5	44.2	1389.6	648.3
Experiment [5]	–	–	1372.4	586.2

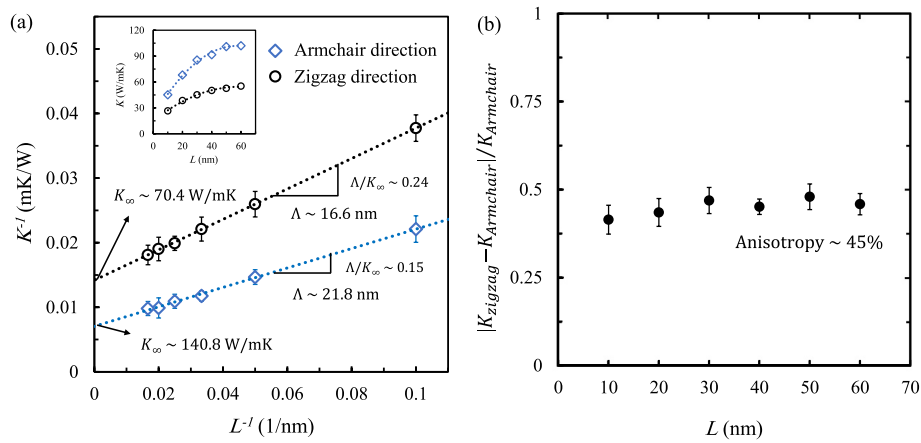


Fig. 2. (a) Thermal conductivity versus the inverse of the length for pristine borophene and (b) Thermal conductivity anisotropy versus the length of borophene nanostructure.

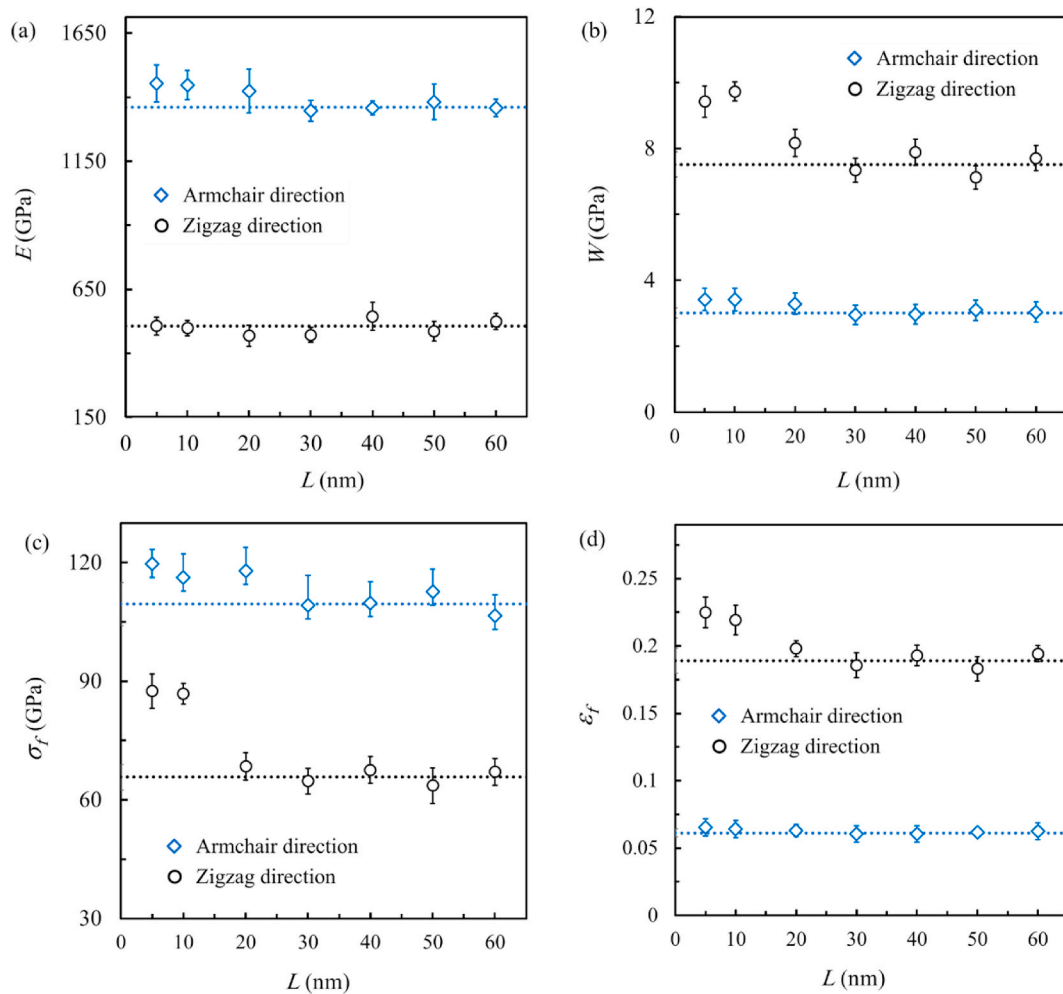


Fig. 3. Size dependency of mechanical properties of pristine borophene: (a) Elastic modulus, (b) Toughness, (c) Fracture stress, and (d) Fracture strain/stretchability.

thermal conductivities of porous borophenes have been normalized by the thermal conductivity of the pristine borophene (K_p) in its corresponding direction (*armchair* or *zigzag*). As shown in Fig. 4, thermal conductivities in *zigzag* and *armchair* directions decrease by the increase of porosity, stemming from the stronger anharmonic phonon–phonon scattering at a higher porosity [61]. The numerical results suggest that 30% porosity in borophene results in more than 50% reduction in

thermal conductivity in both directions, an observation that is compatible with those reported in microstructured porous materials [56]. In addition, comparing the results of porous borophene with porous graphene [40] reveals a similarity in the variation of thermal conductivity with porosity; however, thermal conductivity of borophene is more susceptible to porosity than graphene.

A few correlations have recently been introduced for the thermal

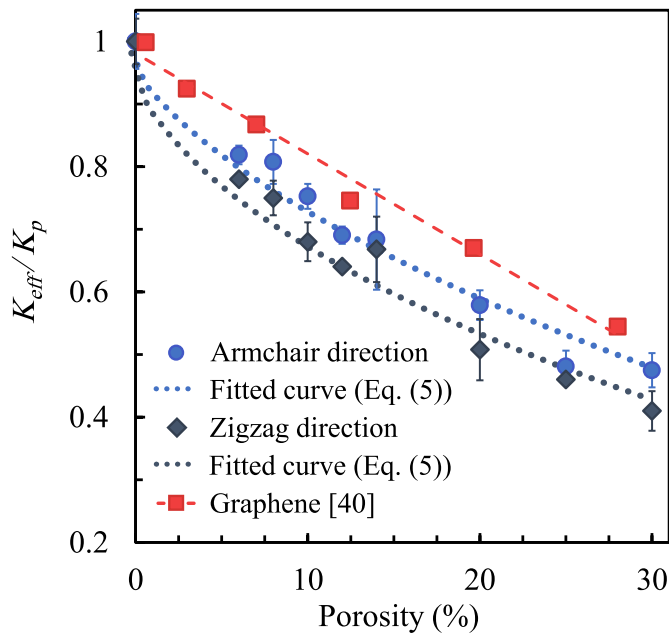


Fig. 4. Effect of porosity on the effective thermal conductivity along armchair and zigzag directions (normalized by the thermal conductivity of pristine form) compared to porous graphenes.

conductivity of porous nanomaterials. In nanomaterials, where the sizes are in the same order of the mean free path, the thermal conductivity is highly dependent on phonon transport and scattering. Effective thermal conductivity can be written as a function of porosity and mean free path for a circular pore as [62,63]:

$$K_{eff} = K_p \frac{(1-P)}{1 + \left(\frac{1}{2} + F\right)P}; \quad F = \frac{\Lambda}{\left(2\sqrt{\frac{AP}{\pi}}\right)} \quad (5)$$

where P represents porosity, Λ is the mean free path of constitutive solid material, and A indicates the planar surface area of the specimen.

Fig. 5 illustrates the NEMD results of a porous borophene sheet containing a circular pore ($P = 8\%$) under a uniaxial tension (with a strain rate of $1 \times 10^{-9} s^{-1}$) along *armchair* and *zigzag* directions. The single circular nanopore is located at the center of the 30×30 nm borophene sheet, while periodic boundary conditions are applied perpendicular to the tensile direction. As found in Fig. 5a, fracture strength in the *armchair* direction is 46% higher than the corresponding value in the *zigzag* direction. This is opposed to the fracture strain/stretchability behavior, where its value in the *zigzag* direction is 120% higher than the *armchair* direction. Similar to the *zigzag* direction, the stress-strain response of circular porous borophene along the *armchair* direction follows a linear elastic regime followed by a nonlinear regime up to the ultimate fracture. Moreover, borophene toughness along the *armchair* direction is found to be less than the *zigzag* direction. The preliminary observation on mechanical properties demonstrates that by introducing a circular pore in the borophene nanostructure, a reduction occurs in the stiffness, strength, and toughness of porous borophene compared to its pristine form (Fig. S2 in Supporting Information). The stretchability to weight (density) ratio also shows a decline in porous borophene. Fig. 5b shows the distribution of von Mises stress when the borophene sheet is stretched in *zigzag* and *armchair* directions. The stress distribution shows the stress localization in a direction perpendicular ($\theta \sim \pm 90^\circ$) to the loading in *zigzag* direction, while the shear bands are found to occur in $\theta \sim \pm 60^\circ$ when porous borophene is stretched in *armchair* direction.

Fig. 6 depicts the effect of porosity on the effective elastic modulus, toughness, tensile fracture strength, and fracture strain/stretchability of

circular porous borophene. Increasing porosity, monotonically decreases all the mechanical properties of porous borophene, showing a linear correlation between the mechanical properties and porosity for $P < 30\%$.

Fig. 7 shows the effect of multiple unit cells in the thermomechanical properties of 12% porous borophene. Due to the size dependency of thermal conductivity, the thermal conductivity of porous borophenes with 12% of porosity increases by the number of cells. As shown in Fig. 7a, the thermal conductivity of porous borophene increases by 48% in both *armchair* and *zigzag* directions for a 150 nm (5L). The thermal conductivity is converged to the finite value of 125.5 W/mK and 67.2 in *armchair* direction and *zigzag* directions, respectively. Fig. 7b evaluates the effect of cell numbers on the elastic modulus of 12% porous borophene. It indicates that the impact of the number of cells on elastic modulus is negligible. Furthermore, the anisotropy of both elastic modulus and thermal conductivity of borophenes are poorly affected by the number of cells.

The 2-Pmmn phase of borophene comes as few-layer material as well as monolayer form [5]. Given that, it is important to study effect of pore engineering on the thermomechanical properties of few-layer borophenes. Here, the effect of circular defect on the thermal and mechanical properties of bilayer and trilayer borophene has been investigated and compared with the monosheet results. Due to the weak van der Waals interactions between boron atoms, no significant change is observed in the mechanical and thermal behavior of porous (Fig. 8) and pristine borophene (Figs. S3 and S4 in Supporting Information). However, as shown in Fig. 8, thermal conductivity and elastic modulus anisotropy of the borophene experience a slight decrease as the number of layers increases. Similar behavior has been observed for the case of pristine borophene (Fig. S5 in Supporting Information).

3.3. Engineering of thermal and mechanical anisotropy

To engineer the inherent anisotropic properties of borophene by pore engineering, we investigate the effect of pore topology and porosity on the thermo-mechanical properties by NEMD simulation and FEM. In specific, we explore the impact of pore size, elongation (aspect ratio $AR = b/a$), and orientation. To quantify the anisotropic thermo-mechanical properties of engineered porous borophene, the following non-dimensional *anisotropic ratio* is defined for each desirable property:

$$\frac{|\Delta S|}{S_{Armchair}} = \frac{|S_{Armchair} - S_{Zigzag}|}{S_{Armchair}} \quad (6)$$

where S can represent thermal conductivity, elastic modulus, toughness, fracture strength, and fracture strain.

The anisotropic ratio for thermal conductivity is shown in Fig. 9 for alternative porosities and pore aspect ratios of porous borophene containing periodic elliptical pore. The thermal conductivity anisotropic ratio of porous borophene remarkably decreases as the pore aspect ratio increases up to a quasi-static isotropic point, $\frac{|\Delta K|}{K_{Armchair}} = 0$, beyond which the thermal conductivity anisotropy ratio increases by a further increase in the pore aspect ratio. Fig. 9 suggests that the pore aspect ratio of the square-isotropy point of thermal conductivity decreases by increasing the porosity of the borophene sheet. For 8% porosity, isotropy point occurs at $AR = 4$, while it happens at $AR = 2.5$ for 12% porosity. The predictions for isotropy point by NEMD and FEM are close; the discrepancy mainly emanates from the size-independency of classical thermoelectricity model used in FEM analysis compared to the size-dependent properties predicted by NEMD simulation. As the porosity increases, the size effect is enhanced in the NEMD simulation, resulting in a higher deviation of FEM and NEMD results.

Fig. 10 shows that the anisotropic ratio for elastic modulus can be controlled by tuning porosity and aspect ratio of the elliptical pore of porous borophene sheets. Opposed to thermal conductivity that shows a 43% anisotropy ratio for pristine borophenes, the corresponding

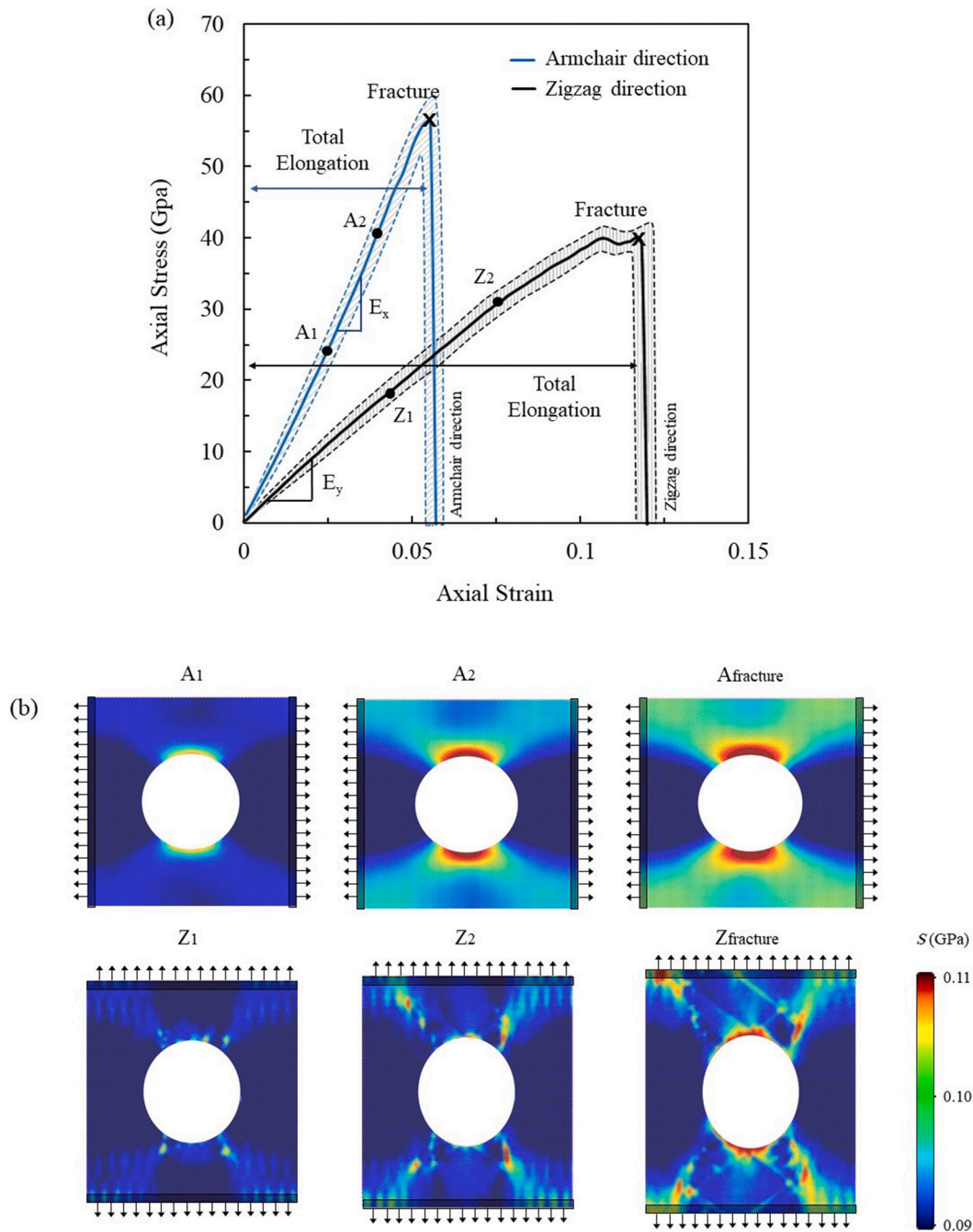


Fig. 5. (a) Stress-strain relation of porous borophene under uniaxial tensile loads in the armchair and zigzag directions and (b) Borophene sheets with a periodic circular pore under tension till rupture, analyzed by NEMD simulation.

anisotropy ratio for elastic modulus is 56%. The aspect ratios in which elastic modulus isotropic point occur are, respectively, 5, 3.8, and 3.5 for 8%, 10%, and 12% porosities. The trend of variation of elastic modulus anisotropic ratio by pore aspect ratio, predicted by FEM, is similar to the NEMD simulation results. Comparing the NEMD and FEM results in Figs. 9 and 10 corroborates the acceptable accuracy of the finite element method for investigating the level of anisotropy of thermo-mechanical properties of engineered porous borophenes.

To better understand the anisotropic thermal conductivity behavior of porous borophene, Fig. 11 illustrates the effect of porosity, pore aspect ratio, and pore orientation (θ) on the effective thermal conductivity by NEMD. Fig. 11 shows that there is no possibility for achieving a square quasi-isotropic state for thermal conductivity for pore orientations between $\theta = 0^\circ$ to $\theta = 60^\circ$. Elliptical pores aligned in an angle

higher than 60° reveal higher feasibility for finding a nanoarchitecture with quasi-isotropic properties. As shown in Fig. 11, the higher the pore aspect ratio, the lower pore angle of the porous borophene sheet associated with isotropic thermal conductivity.

Fig. 12 shows the effect of porosity, aspect ratio, and elongation of elliptical pore on the elastic modulus of porous borophene sheets. Fig. 12a reveals that for $P < 8\%$, there is no possibility for matching elastic modulus in two directions of zigzag and armchair. Fig. 12 shows that the elastic modulus of porous borophene can be equal in both armchair and zigzag directions for aspect ratios higher than 3 ($AR > 3$) and pore orientations higher than 60° ($\theta > 60^\circ$).

Material selection for designing advanced nanomaterials requires evaluation of mechanical properties beyond the elastic regime. To determine the inelastic properties of porous borophenes and the

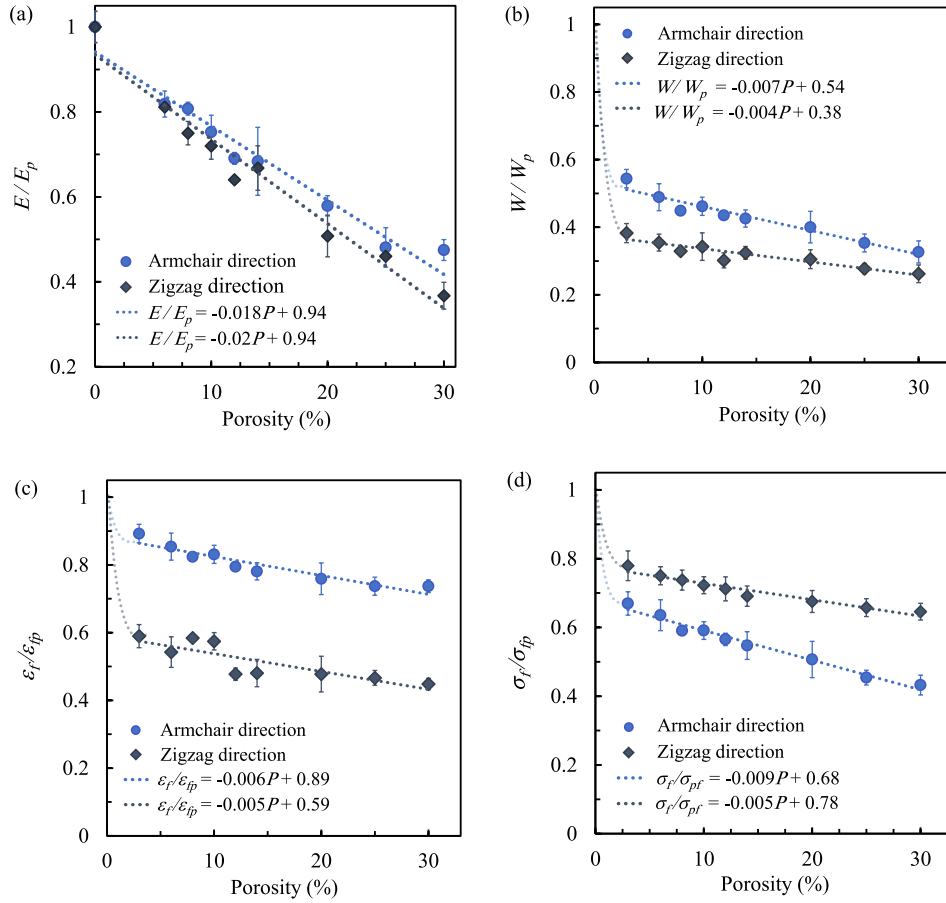


Fig. 6. Mechanical characterization of porous borophene as a function of porosity: (a) Elastic modulus, (b) Toughness, (c) Fracture stress, and (d) Fracture strain/ stretchability (normalized by properties of pristine borophene).

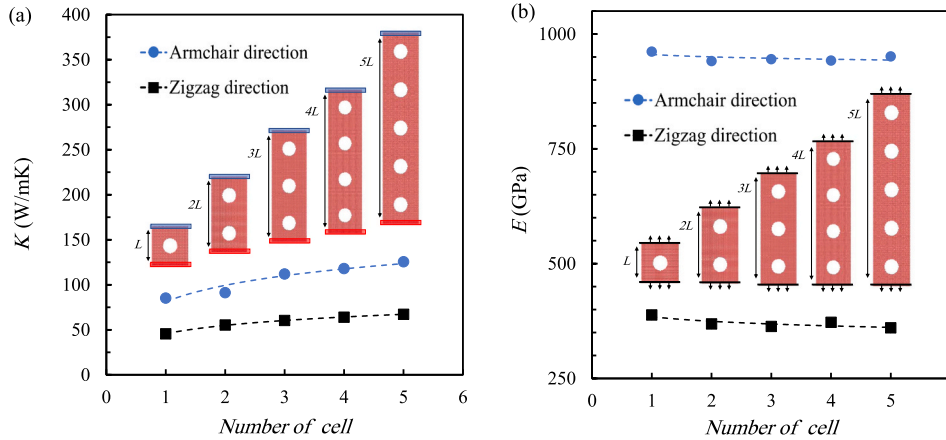


Fig. 7. Effect of unit cell repetition on (a) thermal conductivity and (b) elastic modulus of porous borophene.

associated anisotropy ratios, the effects of porosity, pore aspect ratio, and pore orientation on the effective fracture strain, fracture strength, and toughness are presented in Fig. 13. Pristine borophene shows a 50% difference between the fracture strain in *zigzag* and *armchair* directions. Introducing a circular pore on the borophene nanostructure uniformly reduces the fracture strain in both directions. Fig. 13a shows that the most efficient approach to achieve equal fracture strain in the *armchair* and *zigzag* directions are changing the orientation of elliptical pores. The fracture strength of freestanding borophene is higher in the *zigzag* direction than the *armchair* one. As shown in Fig. 13b, by increasing the

aspect ratio for pores oriented at $\theta = 0^\circ$, anisotropy ratio of fracture strength drastically increases. At one point around $\theta = 60^\circ \pm 15^\circ$, the fracture strength is almost the same in both *zigzag* and *armchair* directions. As the pore rotation goes beyond the quasi-isotropic point, the anisotropic ratio of fracture strength enhances again by increasing the pore elongation. Fig. 13c demonstrates that the borophene toughness anisotropy is lower than that of the fracture stress and fracture strain. It is found that possibility of reaching the quasi-isotropic point of toughness increases for higher porosities ($P > 10\%$).

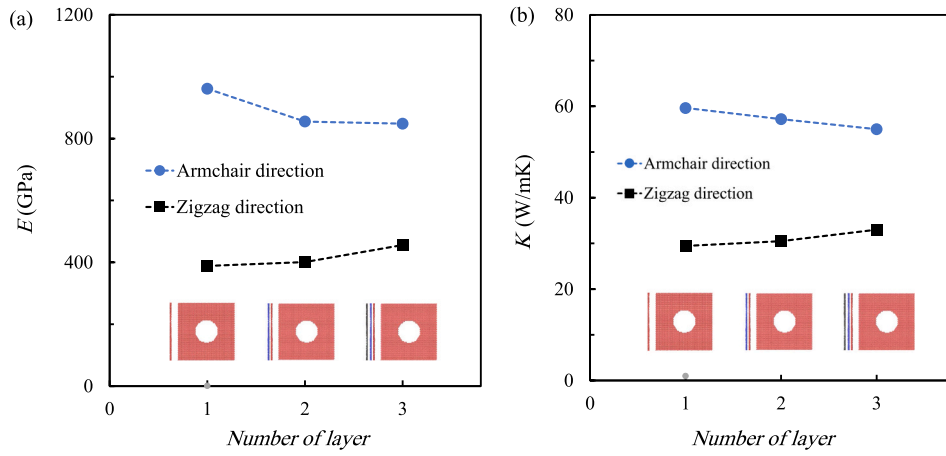


Fig. 8. (a) Elastic modulus and (b) thermal conductivity of porous few-layer borophene ($P = 12\%$) as a function of the number of layers.

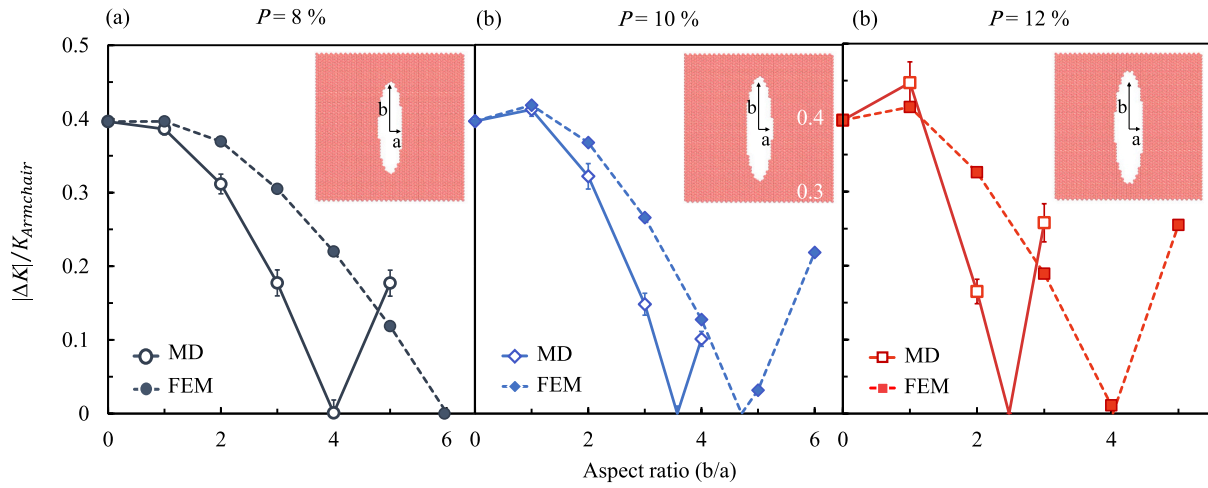


Fig. 9. Thermal conductivity anisotropy ratio of engineered porous borophenes versus the aspect ratio of the elliptical pore determined by NEMD simulation and FEM for: (a) 8%, (b) 10%, and (c) 12% porosities.

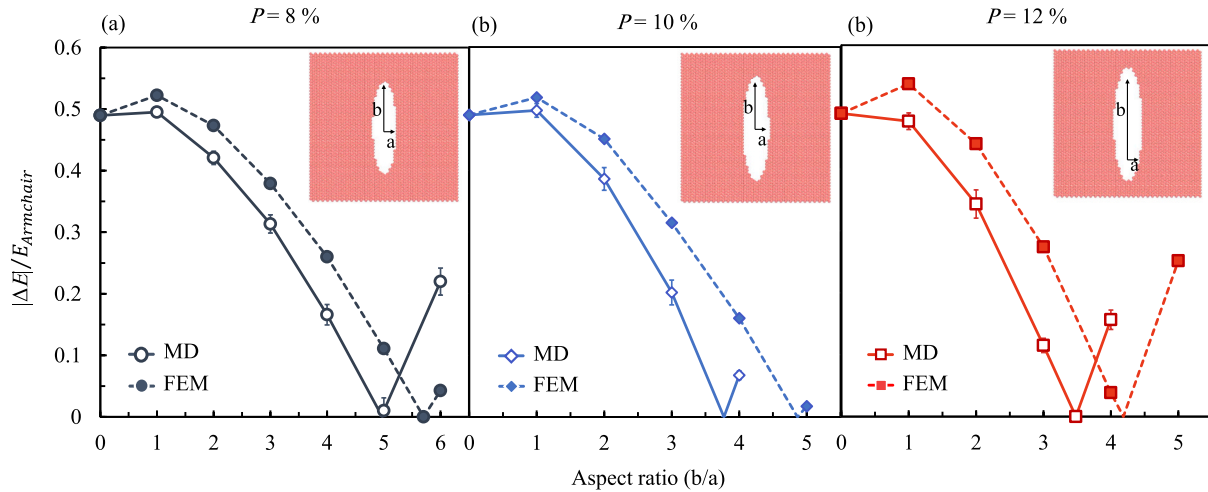


Fig. 10. Elastic modulus anisotropic ratio of engineered porous borophenes versus the aspect ratio of elliptical pore determined by NEMD simulation and FEM for: (a) 8%, (b) 10%, and (c) 12% porosities.

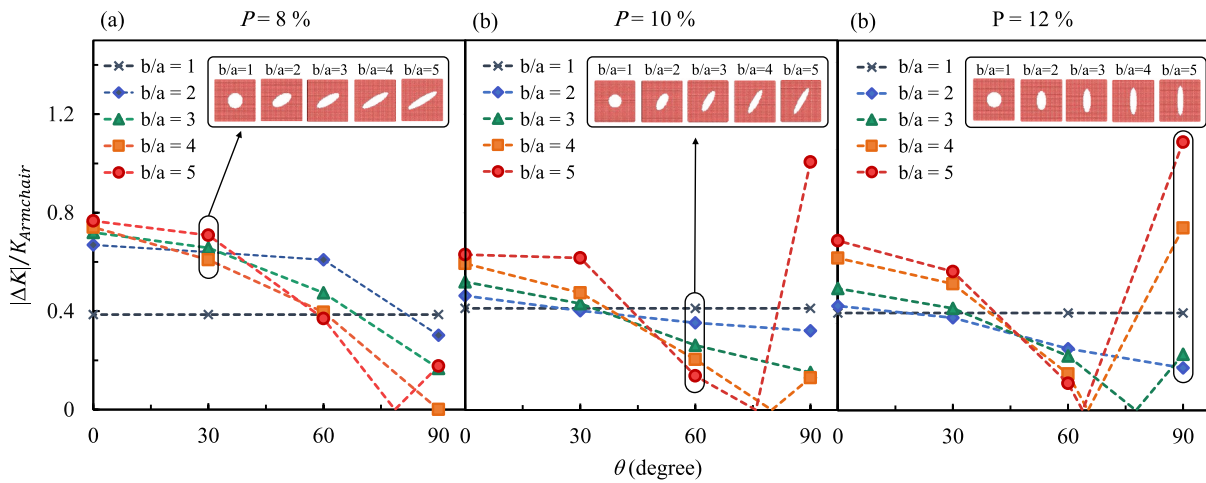


Fig. 11. Thermal conductivity anisotropy ratio of engineered porous borophene versus pore orientation of the elliptical pore determined by FEM for (a) 8%, (b) 10%, and (c) 12% porosities for a wide range of aspect ratio.

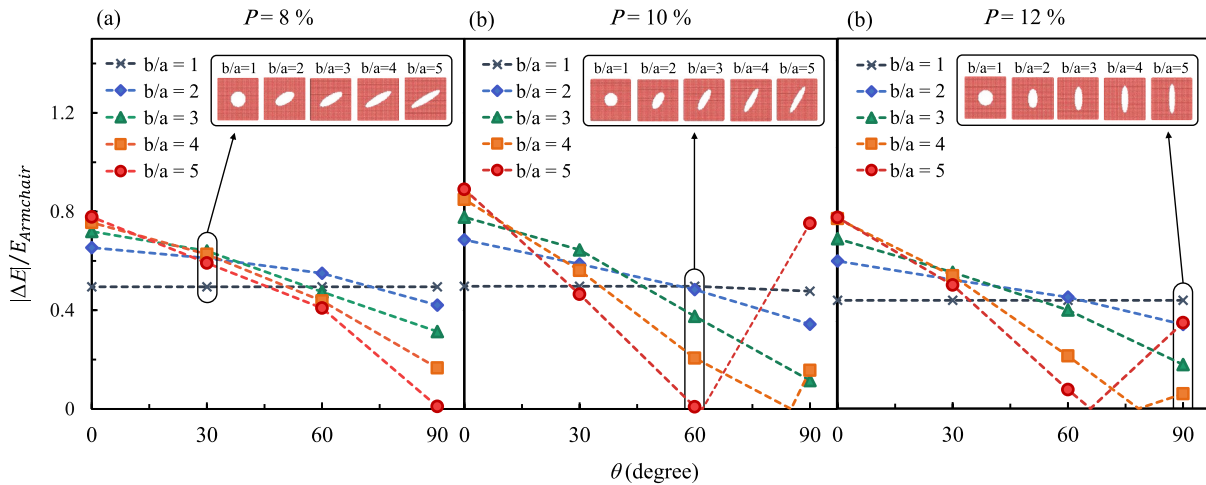


Fig. 12. Elastic modulus anisotropic ratio of engineered porous borophene versus pore orientation of the elliptical pore determined by FEM for: (a) 8%, (b) 10%, and (c) 12% porosities for a wide range of aspect ratio.

4. Concluding remarks

Resorting to non-equilibrium molecular dynamics (NEMD) simulation and finite element modeling (FEM), pore engineering strategy is introduced as an effective approach for controlling the thermal and mechanical properties and the level of anisotropy of multifunctional properties of porous borophenes. Concluding remarks can be summarized as follows:

- It is found that the thermal conductivity of pristine borophene is considerably length-dependent up to $L \sim 60$ nm; however, the length dependence of mechanical properties almost vanishes for lengths greater than $L \sim 30$ nm.
- It is found that 30% porosity in borophene results in more than a 50% reduction in thermal conductivity in both directions. The results reveal that the thermal conductivity of borophenes is more susceptible to porosity than graphenes. A descending linear correlation is also observed between the elastic modulus, fracture strain, fracture strength, and toughness and porosity for $P < 30\%$.
- The effect of multiple unit cells in the thermomechanical properties of porous borophene is also investigated. The thermal conductivity of porous borophene increases by 48% for a 150 nm (5×30 nm (cell size)), while it has no significant effect on the elastic modulus.

- Studying on bilayer and trilayer porous borophene also showed that thermal conductivity and elastic modulus anisotropy experience a slight decrease as the number of layers increases.
- The role of pore topology, pore aspect ratio (elongation), and pore orientation in the engineering of anisotropy ratios of thermo-mechanical properties of porous borophene have been studied. It is found that the lattice thermal conductivity and mechanical anisotropy of porous borophene can be engineered by changing the pore nanoarchitecture. Moreover, the elliptical pores elongated in the zigzag direction can facilitate achieving a quasi-isotropic material property in porous borophenes.
- Comparing the NEMD and FEM results show that both methods are capable of predicting a similar trend for the effect of topological features on the anisotropic thermo-mechanical properties of borophene. However, NEMD results in more accurate predictions as it takes into account the impact of size-dependency for the evaluation of thermo-mechanical properties.

The results of the current study shed light on the unparalleled potential of defect/pore engineering for tuning the thermo-mechanical properties of borophene to program its functionality for applications as stretchable electronics and supercapacitor devices. Pore engineering promises an effective strategy for tuning thermo-mechanical properties

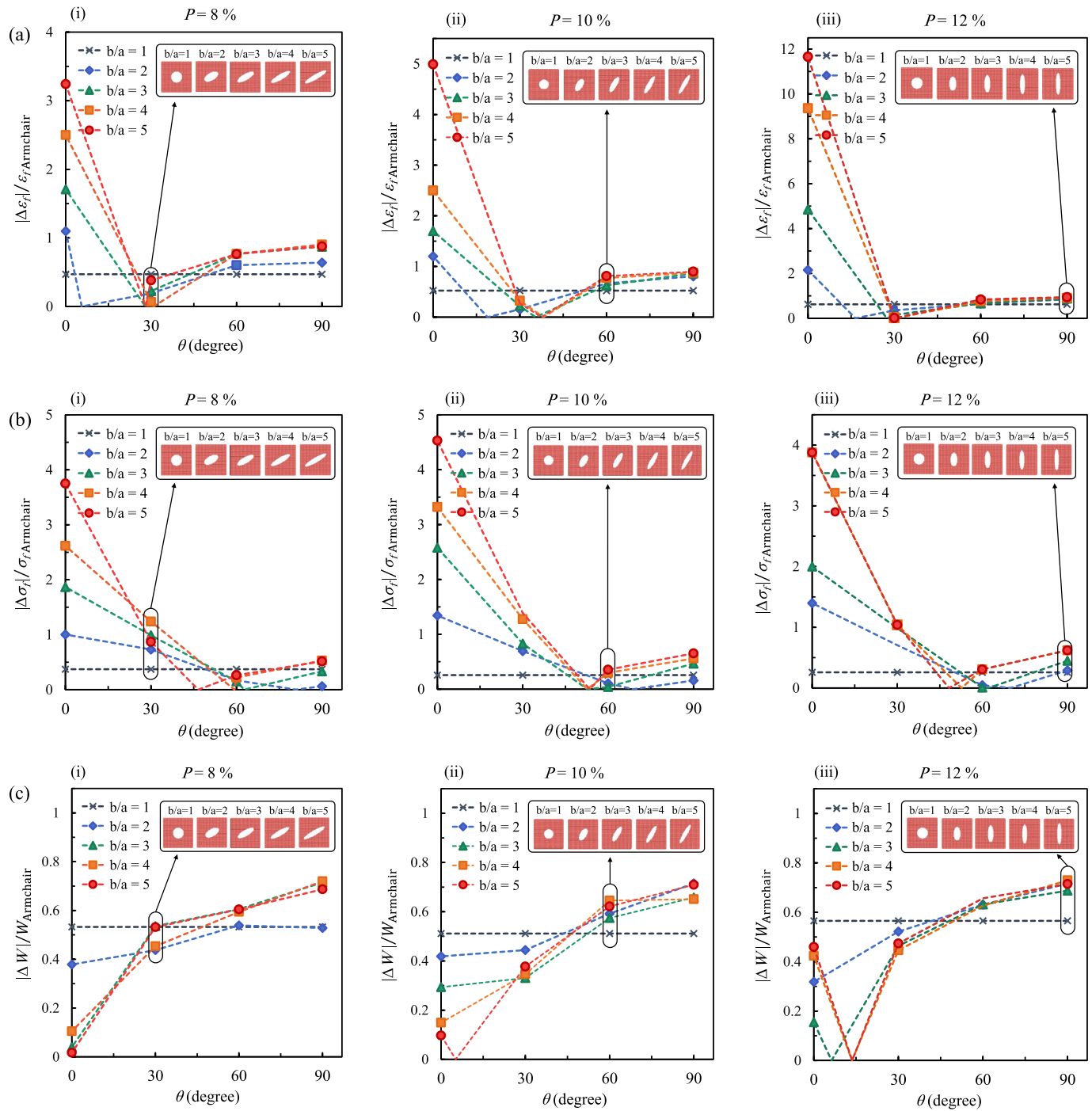


Fig. 13. Anisotropic ratios for: (a) Fracture strain/stretchability, (b) Fracture strength, and (c) Toughness of engineered porous borophenes for alternative aspect ratios, porosity, and pore orientation.

to achieve a quasi-isotropic property in two-dimensional nanostructures, including borophenes and graphenes.

CRedit authorship contribution statement

S. Arabha: Conceptualization, Software, Visualization, Investigation, Methodology, Writing - original draft. **A.H. Akbarzadeh:** Supervision, Conceptualization, Methodology, Writing - review & editing. **A. Rajabpour:** Supervision, Conceptualization, Writing - review & editing, Methodology, Validation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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